

1. Which of the following statements is/are correct regarding RTL synthesis of an always block in an RTL model written in Verilog?

1 Point

- A. An always block can never be synthesized to a combinational logic.
- B. An always block can sometimes be synthesized to a latch.
- C. An always block can sometimes be synthesized to a flip-flop.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☒ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only B, C

2. Which of the following statements is/are correct regarding RTL synthesis of an RTL model written in Verilog?

1 Point

- A. The initial block is typically ignored by an RTL synthesis tool.
- B. The delay of 10 specified in the statement: `reg2 <= #10 reg1` will be ignored by the RTL synthesis tool.
- C. System functions such as `$display` and `$monitor` are typically synthesized to latches.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

3. What is the expected RTL synthesis output of the following piece of RTL code?

1 Point

```
module top(a, b, c, s, en, out1, out2);  
  input a, b, c, s, en;  
  output out1, out2;  
  reg out1, out2;
```

```
  always @(*) begin  
    if (s==1'b0)  
      out1 = a;  
    else  
      out1 = b;  
    end
```

```
  always @(*) begin  
    if (en==1'b1)  
      out2 = c;  
    end  
endmodule
```

- ☐ Both **out1** and **out2** driven by latches
- ☐ Both **out1** and **out2** driven by flip-flops
- ☐ Both **out1** and **out2** driven by 2-to-1 multiplexers
- ☐ **out1** driven by a latch and **out2** driven by a 2-to-1 multiplexer
- ☐ **out1** driven by a latch and **out2** driven by a flip-flop
- ☐ **out1** driven by a flip-flop and **out2** driven by a 2-to-1 multiplexer
- ☐ **out1** driven by a flip-flop and **out2** driven by a latch
- ☒ **out1** driven by a 2-to-1 multiplexer and **out2** driven by a latch
- ☐ **out1** driven by a 2-to-1 multiplexer and **out2** driven by a flip-flop

Yes, the answer is correct. Score: 1

Accepted answer is **out1** driven by a 2-to-1 multiplexer and **out2** driven by a latch

4. What will be the typical RTL synthesis result of the following piece of RTL code?

1 Point

```
module top1(in1, clk, out1);  
  input in1, clk;  
  output out1;  
  reg reg1, reg2, reg3, out1;
```

```
  always @(posedge clk)  
  begin  
    reg1 = in1;  
    reg2 = reg1;  
    reg3 = reg2;  
    out1 = reg3;  
  end  
endmodule
```

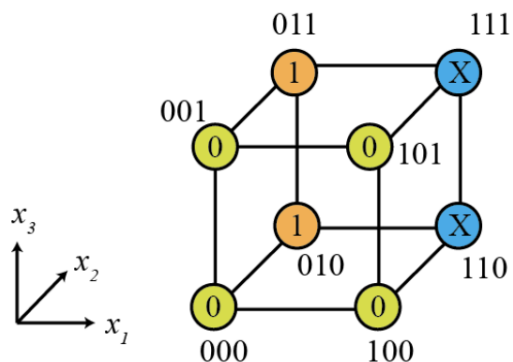
- ☐ One 2-to-1 multiplexer
- ☐ One latch
- ☒ One flip-flop
- ☐ 3-bit shift register
- ☐ 4-bit shift register

Yes, the answer is correct. Score: 1

Accepted answer is One flip-flop

5. Consider the Boolean function of three Boolean variables x_1, x_2 , and x_3 . The hypercube representation of the function is shown below.

1 Point



Which of the following statements is/are correct for the above function?

- A. x_1' is a prime implicant
- B. x_2 is a prime implicant
- C. $x_2'x_3$ is a prime implicant

- ☐ only A
- ☒ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct.
- ☐ No statement is correct.

Yes, the answer is correct. Score: 1

Accepted answer is only B

6. Which of the following statements is/are correct about implicants and prime implicants?

1 Point

- A. An implicant of a function can include the corners of the OFF-set of the function.
- B. An implicant of a function can include corners of the DC-set of the function.
- C. A prime implicant is an implicant of a function that is not covered by any other implicant of that function.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☒ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only B, C

7. Which of the following statements is/are correct about the heuristic minimizer in logic optimization?

1 Point

- A. A heuristic minimizer may not produce the exact minimum cover for a Boolean function.
- B. A heuristic minimizer starts with an initial cover and iteratively improves the solution by applying operators to it.
- C. A heuristic minimizer is typically slower than the exact minimizer, especially for large problems.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is: only A, B

8. Which of the following statements is/are correct about optimizations done in RTL synthesis?

1 Point

- A. The primary objective of resource sharing is to reduce the area.
- B. Resource sharing can sometimes lead to an increase in the delay of the critical path.
- C. Unsharing resources can sometimes reduce the delay of the critical path.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☒ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is: All statements are correct

9. Which of the following statements is/are correct about synthesis of a for loop in Verilog?

1 Point

- A.A for loop is not synthesizable if the number of iterations cannot be statically known or derived by the synthesis tool.
B.A for loop can be unrolled during RTL synthesis if the number of iterations is fixed or can be derived statically by the tool.
C.A for loop is never synthesized by an RTL synthesis tool.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

10. Consider the following statements about the tools ICARUS, GTKWave, and COVERED used in simulation-based verification.

1 Point

- A.We can use the ICARUS tool to write a VCD file containing simulation results for a given Verilog design and testbench.
B.We can use GTKWave to display the results from a VCD file.
C.The COVERED tool is used to obtain a maximal cover for a Boolean function.

Which of the above statements is/are correct?

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, B